Fork, Explore, Commit: OS Primitives for Agentic Exploration

Cong Wang[1], Yusheng Zheng[2]

[1]Multikernel Technologies, Inc. · [2]UC Santa Cruz

Agentic OS Workshop, ASPLOS 2026

Problem: Agents Explore, But Have Side Effects

Agentic Exploration

- AI agents (SWE-Agent, OpenHands, Claude Code) execute **multi-step tasks**
- Shell commands, file edits, package installs → **irreversible side effects**
- Increasingly exploring **multiple paths in parallel**: Tree-of-Thoughts, Reflexion, Best-of-N
- Need to **isolate** each path, **commit** the winner, **discard** the rest

What Do Agents Use Today?

Approach	Limitation
Git stashing	Misses untracked files
Temp directories	No atomic commit

Example

> A software engineering agent tries 3 candidate fixes simultaneously. Only the one that passes tests should be committed — the rest must be cleanly discarded.

The Gap

> **None** captures all filesystem modifications with atomic commit/rollback.

- **LangChain & AutoGPT**: git stashing or temp directories
- **Claude Code**: per-file snapshots, can't capture `npm install` side effects
- **All**: no parallel branching, no sibling invalidation

Six Requirements for Agentic Exploration

#	Requirement	Why
R1	**Isolated parallel execution**	Concurrent paths modify same files
R2	**Atomic commit + single-winner**	Apply winner's changes, invalidate siblings
R3	**Hierarchical nesting**	Tree-of-Thoughts explores sub-variants
R4	**Complete filesystem coverage**	Must capture *all* modifications, not just tracked files
R5	**Lightweight, unprivileged, portable**	Sub-ms creation, no root, any FS (ext4, XFS, NFS...)
R6	**Process coordination**	Reliable termination, sibling isolation

No existing OS mechanism satisfies all six requirements.

Why Existing Mechanisms Fall Short

Filesystem Branching

Feature	OverlayFS	Btrfs/ZFS	DM-Snap	**BranchFS**
Portable FS	✓	✗	✓	✓
Nested branches	✗*	✓	✗	✓
Commit-to-parent	✗	✗	✗	✓
Sibling invalidation	✗	✗	✗	✓
No root	✗	✗	✗	✓

Process Management

Feature	pgrp	cgroup	PID ns	**branch()**
Reliable termination	✗	✓	✓	✓
No escape	✗	✓	✓	✓
Sibling isolation	✗	✗	✓	✓
No setup / No root	✓	✗	✗	✓
No PID 1 overhead	✓	✓	✗	✓

- **OverlayFS**: no commit semantics, requires root. **Btrfs/ZFS**: FS-specific. **DM-Snap**: O(depth) latency
- **Process groups**: escapable via `setsid()`. **Cgroups**: need setup + root. **PID ns**: PID 1 overhead
- Composing these in **userspace** creates race windows and fragile error handling

Our Solution: The Branch Context

Definition & Lifecycle

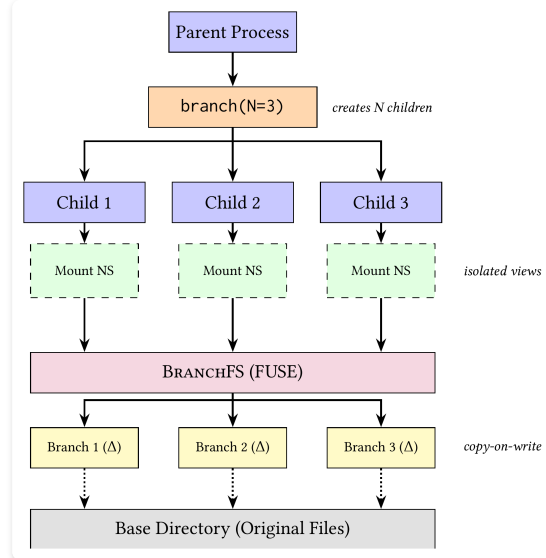
A **branch context** = CoW filesystem view (Δ_i) + confined process group

```
Fork ──▶ Explore ──▶ Commit (winner)
                  └──▶ Abort  (losers)
```

Four Core Properties:

1. **Frozen origin** — parent read-only; no merge conflicts
2. **Parallel isolated execution** — N contexts, fully isolated
3. **First-commit-wins** — siblings auto-invalidated
4. **Nestable** — sub-contexts form exploration tree

Architecture



branch() = process coordination; **BranchFS** = filesystem CoW

BranchFS: FUSE-Based Copy-on-Write

File-Level CoW (~3,400 lines of Rust)

- First write → **copy entire file** to branch's delta layer
- Subsequent reads/writes served from **delta copy**
- Unmodified files resolved by walking the **branch chain**

Branch Chain Resolution

1. Check current branch's delta
2. Walk ancestor branches in order
3. Fall back to base directory
4. **Tombstone markers** prevent deleted files from reappearing

Key Design Properties

O(1) branch creation — just create a delta directory

No root privileges — runs entirely as a FUSE daemon

Portable — works on ext4, XFS, NFS, any FS

Commit & Abort

- **Commit**: copy delta to parent, increment **epoch counter** → invalidates all siblings
- **Abort**: discard delta layer → **near-zero cost**
- Cost proportional to **modification size**, not total FS size

BranchContext: Python Integration Library

7 Exploration Patterns

Pattern	Strategy
Speculate	Race N candidates, first success wins
BestOfN	Run N, commit highest-scoring
Reflexion	Sequential retry with feedback
TreeOfThoughts	Hierarchical nested branches
BeamSearch	Keep top-K at each depth
Tournament	Pairwise elimination via judge
Cascaded	Start with 1, fan out on failure

Usage Example

```python
ctx = BranchContext("/workspace")

# Best-of-N: try 3 fixes, commit best
results = ctx.best_of_n(
    n=3,
    task=lambda b: try_fix(b),
    score=lambda b: run_tests(b)
)
# Winner automatically committed
```

- Each pattern manages **branch lifecycle** + **process isolation** internally
- Agent code only supplies **per-branch task logic**

github.com/multikernel/branching

The branch() Syscall

Why a Kernel Primitive?

Setting up cgroups + PID ns + mount ns + FS branches in **userspace**:

- **Multi-step** with race windows between steps
- A process can fork between cgroup creation and migration
- **Error-prone** cleanup on partial failure

`branch()` composes all atomically in a **single call** with kernel-side cleanup on failure.

Three Operations

- `BR_CREATE` — fork N children, each in its own branch
- `BR_COMMIT` — apply FS changes, terminate siblings
- `BR_ABORT` — discard changes, terminate self

Interface

```
long branch(int op,
            union branch_attr *attr,
            size_t size);
```

Composable Flags

Flag	Effect
`BR_FS`	Mount namespace + FS branch
`BR_MEMORY`	Page-table CoW
`BR_ISOLATE`	Signal/ptrace barriers
`BR_CLOSE_FDS`	Close inherited FDs

- FS-agnostic: generic `FS_IOC_BRANCH_*` ioctls
- Adding a new branching FS = implement 3 ioctls

Preliminary Evaluation

Branch Creation — O(1)

Base Size	Latency
100 files	292 µs
1,000 files	317 µs
10,000 files	310 µs

Independent of base size.

Commit & Abort

Mod. Size	Commit	Abort
1 KB	317 µs	315 µs
100 KB	514 µs	365 µs
1 MB	2.1 ms	890 µs

Proportional to mod size.

I/O Throughput

Mode	Read
Native	8.8 GB/s
FUSE	1.7 GB/s
Passthrough	**7.2 GB/s**

82% native with passthrough.

Agent workloads dominated by **LLM API latency** (100 ms – 10 s) — BranchFS I/O overhead is **negligible**.

Hardware: AMD Ryzen 5 5500U, 8 GB DDR4, NVMe SSD. Median of 10 trials. BranchFS: ~3,400 lines of Rust, FUSE 3.

Related Work & Future Directions

Related Work

- **Containers/gVisor/Dune**: general containment, not branching
- **Speculator**: sequential speculation for distributed FS — we do parallel N-way with first-commit-wins
- **TxOS**: flat, short-lived OS transactions — we're nestable, long-lived, userspace FUSE
- **lwCs/Capsicum**: different isolation granularity

Current Limitations

- **External side effects** (network, IPC) not rolled back
- **Single-winner** only — no multi-branch merge
- File-level CoW: symlinks, hardlinks unsupported
- **BR_MEMORY** not yet implemented

Future Directions

Effect gating — buffer network/IPC until commit; agent gateways provide interposition points

Multi-branch merge — union non-overlapping changes; conflict detection

Roadmap — prototype `branch()` on Linux 6.19; BR_FS + BR_ISOLATE first

Broader use — `n_branches=1` → try-and-rollback for package mgmt, system tuning

Key Takeaways

1. **Agentic exploration** is a first-class OS concern: agents need isolated, parallel workspaces with atomic commit/rollback
2. **Branch context** = new OS abstraction with fork/explore/commit lifecycle and first-commit-wins resolution
3. **BranchFS** = working FUSE implementation: O(1) creation, no root, portable across filesystems
4. **BranchContext** = Python library with 7 ready-to-use exploration patterns
5. **branch() syscall** = proposed kernel primitive for atomic process + FS coordination

BranchFS: **github.com/multikernel/branchfs**

BranchContext: **github.com/multikernel/branching**

Thank You — Questions?

Cong Wang — cwang@multikernel.io

Yusheng Zheng — yzhen165@ucsc.edu

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